10MAA Revision Examination: Paper 2

Worked marking schedule

Award method marks for valid working, including equivalent methods. FT means follow through a previous error when the later method is valid. CAO means correct answer only, with no working needed for that mark. Accept exact equivalents unless a question specifies a form or accuracy. Do not award a mark twice for the same step.

Number: 12 marks

1. Original price $= \dfrac{1380}{1.15} = \$1200$.

1 for multiplier 1.15, 1 for division, 1 for the price.

2. $2400(1.04)^3 = \$2699.6736$, so $\$2699.67$.

1 for multiplier, 1 for compounding three times, 1 for balance.

3. Let $x = 0.363636\ldots$. Then $100x = 36.363636\ldots$, so $99x = 36$ and $x = \dfrac{4}{11}$.

1 for setting up a repeating decimal, 1 for subtraction, 1 for simplest fraction.

4. 15 minutes is $\dfrac{1}{4}$ hour. $3.2 \times 10^6 \div 4 = 8 \times 10^5$ records.

1 for time conversion, 1 for calculation, 1 for standard form.

Algebra: 32 marks

1. $3x = 18$, so $x = 6$.

1 for collecting terms, 1 for 6.

2. $6t - 3 > 4t + 7$, so $t > 5$.

1 for collecting terms, 1 for the correct inequality.

3. $2a^4b$.

1 for coefficient 2, 1 for both indices.

4. $x^2 - 2x - 15$.

1 for expansion, 1 for simplification.

5. $6(x - 2)(x + 2)$.

1 for common factor, 1 for complete factorisation.

6. $\dfrac{(x-3)(x+3)}{(x+2)(x+3)} = \dfrac{x-3}{x+2}$; $x \neq -3, -2$.

1 for factorisation, 1 for cancellation, 1 for both exclusions.

7. $h = \dfrac{A}{2\pi r} - r$.

1 for dividing by $2\pi r$, 1 for isolating h.

8. Let l and a be Linux and AI stickers. $l + a = 18$, $2l + 3a = 43$. Thus $a = 7$, $l = 11$.

1 per equation, 1 for solving, 1 for both counts identified.

9. $(x - 5)(x - 2) = 0$; $x = 2$ or 5.

1 for factors, 1 for each solution.

10(a). $\dfrac{120}{x} - \dfrac{120}{x+4} = 1$.

1 for both time expressions, 1 for equation.

10(b). $x(x + 4) = 480$; $(x + 24)(x - 20) = 0$. $x = 20$; reject -24 because a processing rate must be positive.

1 for quadratic, 1 for roots, 1 for 20 with rejection reason. FT an appropriate equation.

11. First differences are $5, 7, 9, 11$, with second difference 2. The quadratic coefficient is 1; $n^2 + 2n$ gives the terms.

1 for second difference, 1 for quadratic coefficient, 1 for $n^2 + 2n$.

12. $x^2 - 6x + 2 = (x - 3)^2 - 7$.

1 for $(x - 3)^2$, 1 for -7.

Statistics and probability: 26 marks

1(a). Median $= 7$ s; mean $= \dfrac{56}{7} = 8$ s.

1 for median, 1 for total 56, 1 for mean.

1(b). Median remains 7 s. The middle, fourth value does not change.

1 for median, 1 for explanation.

2(a). Box from 6 to 11, median at 8, whiskers to 4 and 16.

1 for labelled scale, 1 for box and median, 1 for whiskers.

2(b). $11 - 6 = 5$ s. It measures the spread of the middle 50% of the data.

1 for 5 s, 1 for interpretation.

3(a). $50 - 5 = 45$ use at least one. Both $= 30 + 27 - 45 = 12$.

1 for 45, 1 for inclusion-exclusion, 1 for 12.

3(b). $\dfrac{12}{30} = \dfrac{2}{5}$.

1 for denominator 30, 1 for result. FT (a).

4(a). $\dfrac{3}{8} \times \dfrac{2}{7} = \dfrac{3}{28}$.

1 for product, 1 for answer.

4(b). $\dfrac{3}{8} \times \dfrac{5}{7} + \dfrac{5}{8} \times \dfrac{3}{7} = \dfrac{15}{28}$.

1 for each path, 1 for sum.

4(c). Both orders in (b) have probability $\dfrac{15}{56}$, so $\dfrac{\frac{15}{56}}{\frac{30}{56}} = \dfrac{1}{2}$.

1 for conditional method, 1 for answer. FT consistent path probabilities.

5(a). Midpoints are 5, 15, 30, 50. Mean $= \dfrac{4(5) + 8(15) + 12(30) + 6(50)}{30} = \dfrac{800}{30} = 26.7$ s.

1 for midpoints, 1 for weighted total, 1 for mean.

5(b). $\dfrac{12}{20} = 0.6$.

1 CAO.

Trigonometry: 16 marks

1(a). $\sqrt{12^2 + 9^2} = 15$ m.

1 for Pythagoras, 1 for 15.

1(b). $\theta = \tan^{-1}\left(\dfrac{12}{9}\right) = 53.1°$.

1 for ratio, 1 for angle.

2(a). Bearing of A from B is 270°; angle $ABC = 120°$.

1 for reverse direction, 1 for angle.

2(b). $AC^2 = 8^2 + 6^2 - 2(8)(6)\cos 120° = 148$; $AC = 12.2$ km.

1 for cosine rule, 1 for substitution, 1 for distance. FT (a).

2(c). $\dfrac{1}{2}(8)(6)\sin 120° = 12\sqrt{3} = 20.8$ km².

1 for area formula, 1 for substitution, 1 for area. FT (a).

3(a). Smooth sine curve through $(0°, 0)$, $(90°, 1)$, $(180°, 0)$, $(270°, -1)$ and $(360°, 0)$.

1 for labelled axes and scale, 1 for intercepts, 1 for turning points and shape.

3(b). $\theta = 30°$ or $150°$.

1 for both answers.

Measurement: 15 marks

1. Length scale factor $\dfrac{15}{6} = 2.5$. Volume $= 40(2.5)^3 = 625\text{cm}^3$.

1 for length factor, 1 for cubing it, 1 for volume.

2(a). $\pi(0.4)^2(1.2) = 0.192\pi = 0.603$ m³.

1 for formula and substitution, 1 for volume.

2(b). $0.192\pi \times 1000 = 603$ litres.

1 for conversion factor, 1 for result. FT unrounded (a).

3. $20(12) - 2\pi(3)^2 = 240 - 18\pi = 183$ cm² to nearest cm².

1 for rectangle, 1 for both circles, 1 for answer.

4(a). $14.5 \le l < 15.5$ cm.

1 per bound, including correct inequality.

4(b). Upper width bound $= 8.5$ cm. Supremum $= 15.5 \times 8.5 = 131.75$ cm², not attained.

1 for width bound, 1 for product, 1 for correct value.

Coordinate geometry: 11 marks

1(a). $m = \dfrac{7 - 3}{4 - (-2)} = \dfrac{2}{3}$.

1 for method, 1 for gradient.

1(b). Midpoint $(1, 5)$.

1 CAO.

1(c). $\sqrt{6^2 + 4^2} = \sqrt{52} = 2\sqrt{13}$.

1 for distance formula, 1 for exact length.

1(d). Perpendicular gradient $-\frac{3}{2}$; $y-5=-\frac{3}{2}(x-1)$; $y=-\frac{3}{2}x+\frac{13}{2}$.

1 for perpendicular gradient, 1 for midpoint substitution, 1 for rearrangement, 1 for equation. FT earlier results.

2. $2(10)-23=-3$, so the point lies on the line.

1 for substitution, 1 for conclusion.

Quadratic functions: 20 marks

1(a). Values: $5, 0, -3, -4, -3, 0$.

1 for at least three correct, 1 for all six.

1(b). Plot the six values and join with a smooth parabola.

1 for scale, 1 for points, 1 for smooth curve. FT (a).

1(c). Turning point $(1, -4)$.

1 for each coordinate.

1(d). $(x-3)(x+1)=0$; $x=3$ or -1.

1 for factorisation, 1 for both roots.

1(e). Translation 1 unit right and 4 units down.

1 per direction and distance.

2(a). $h=1\times 7=7$ m.

1 CAO.

2(b). $t=7$ s.

1 CAO.

2(c). Roots are -1 and 7, with midpoint $t=3$; $h=4\times 4=16$ m.

1 for symmetry, 1 for time, 1 for height.

2(d). $(t+1)(7-t)=12$ gives $t^2-6t+5=0=(t-1)(t-5)$. Times are 1 s and 5 s.

1 for equation, 1 for quadratic, 1 for factorisation, 1 for both times.

Geometry: 20 marks

1. Base angles are equal, so angle $ABC=\dfrac{180°-38°}{2}=71°$.

1 for equal base angles, 1 for angle.

2(a). Angle $PSR=180°-112°=68°$. Opposite angles of a cyclic quadrilateral sum to 180°.

1 for reason, 1 for angle.

2(b). Angle $QRS=180°-74°=106°$ (opposite cyclic angles). Angle $QRT=180°-106°=74°$ (straight line).

1 per justified step.

3. Draw $AB=6$ cm. With the same compass radius greater than 3 cm, draw arcs centred on A and B above and below the segment. Join the two intersections to form the perpendicular bisector.

1 for correctly spaced points, 1 for matching arcs, 1 for the line through their intersections.

4(a). $\overrightarrow{MN}=\boldsymbol{q}-\boldsymbol{p}$.

1 CAO.

4(b). $\overrightarrow{PQ}=\frac{1}{2}\boldsymbol{q}-\frac{1}{2}\boldsymbol{p}$.

1 for each position vector, 1 for subtraction.

4(c). $\overrightarrow{PQ}=\frac{1}{2}(\boldsymbol{q}-\boldsymbol{p})=\frac{1}{2}\overrightarrow{MN}$. As one vector is a positive scalar multiple of the other, the segments are parallel and $PQ=\frac{1}{2}MN$.

1 for scalar relation, 1 for parallel conclusion, 1 for length conclusion. FT (a) and (b) if consistent.

5(a). $\overrightarrow{PR}=\frac{1}{3}\boldsymbol{p}+\frac{1}{3}\boldsymbol{q}-\frac{1}{2}\boldsymbol{p}=-\frac{1}{6}\boldsymbol{p}+\frac{1}{3}\boldsymbol{q}$.

1 for subtracting $\overrightarrow{OP}$, 1 for result.

5(b). $\overrightarrow{PN}=\boldsymbol{q}-\frac{1}{2}\boldsymbol{p}$, so $\overrightarrow{PR}=\frac{1}{3}\overrightarrow{PN}$. Starting at P, this puts R on segment PN, one third of the way to N. Hence $PR:RN=1:2$.

1 for $\overrightarrow{PN}$, 1 for the scalar relation and position on segment, 1 for ratio. FT consistent (a).